GEK-83763

DRIVE SYSTEM INSTRUCTIONS

μdrill™ 1500
RIG TROUBLESHOOTING GUIDE

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

μdrill™ 1500
RIG TROUBLESHOOTING GUIDE

GEK-83763

ISSUE DATE: September, 1981

The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, U.S.A.

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HOW TO USE THIS GUIDE

This manual contains troubleshooting information for the u-drill(TM) 1500 rig drive system and consists of four sections: General Information, Initial Problem Determination, Drilling Operation Troubleshooting Guide, and Power Generation Troubleshooting Guide. Section 1, General Information, contains troubleshooting hints, preventive maintenance procedures, and information to help you locate rig drive system subassemblies. Section 2, Initial Problem Determination, is used as the first step in troubleshooting; this section helps you to determine the general area of the problem and will refer you to other sections of the book for further troubleshooting. Section 3, Drilling Operation Troubleshooting Guide, contains troubleshooting procedures that relate to drilling operation problems. Section 4, Power Generation Troubleshooting Guide, contains troubleshooting procedures that relate to power generation problems.

We assume that a user of this guide has read the u-drill(TM) 1500 Rig Operation Manual (GEK-73911) and is familiar with basic rig operation and the location and use of all operator controls. You should also have handy the system elementary drawings.

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* *
* WARNING *
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This equipment contains a potential hazard of electrical shock or burn. Only personnel who are adequately trained and thoroughly familiar with the equipment and these instructions should install, operate, or maintain this equipment.

There are several steps involved in using this guide for troubleshooting. Assuming that a problem exists, you would typically follow these steps to troubleshoot the problem:

1. Refer to Section 1, Initial Problem Determination and find the most likely cause of the problem under the column labelled "Apparent Problem". Follow the related instructions under the column labelled "Action to Take". If there are several possible causes of the problem, pick the most likely cause to troubleshoot first; if that doesn't solve the problem, go back and pick the next most likely cause. The best starting point for some problems may not be certain. If one item does not resolve your problem, review the list again for another possibility. Refer to action item 16 in the list if none of the items seems to cover your specific problem.

2. The "Action to Take" column will typically refer you to one of three places:
   a. another action item (within the table),
   b. the Drilling Operation Troubleshooting Guide, or
   c. the Power Generation Troubleshooting Guide.

3. If you are referred to the Drilling Operation Troubleshooting Guide, you should:
    a. start at the beginning of the guide to identify the problem area,
    b. then refer to the applicable troubleshooting procedure for that problem area,
    c. then refer to the applicable Drilling Operation action item called out by the troubleshooting procedure to correct the problem.

4. If you are referred to the Power Generation Troubleshooting Guide, you should:
    a. start at the beginning of the guide to identify the problem area and applicable Power Generation action item for that problem,
    b. then refer to the action item to correct the problem.

SECTION 1
GENERAL INFORMATION

TROUBLESHOOTING HINTS

Before attempting to troubleshoot the rig equipment using this guide, be sure that:

1. You are familiar with the equipment, the location and operation of each device, and the contents of this troubleshooting guide,
2. You have the elementary diagram handy (this is the most valuable tool you can use in troubleshooting),
3. You have any necessary tools or test equipment handy, and
4. You have on hand adequate spare parts to replace failed components.

During troubleshooting, be careful not to get the equipment out of adjustment. Most rig problems are caused by loose connections, broken wires, or failed components and can be corrected without making additional adjustments to the equipment. Changing the setting of a potentiometer on a board can necessitate a retuning of the entire rig drive system, which involves considerable time and effort.

If changing a printed circuit board does not correct the problem, replace the original board in the card cage. Be careful not to cause damage when removing and replacing boards in card cages. Do not force a board into its slot, since this can damage the board and/or the backplane.

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* *
* CAUTION *
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Do not insert or remove a board without first turning off power. Failure to do so may result in damage to circuitry.

PREVENTIVE MAINTENANCE

Many problems that require troubleshooting can be avoided by establishing a regular preventive maintenance program. A preventive maintenance program can help reduce costly downtime and prevent intermittent problems that interfere with rig operations.

****
* *
* WARNING *
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This equipment contains a potential hazard of electrical shock or burn. Remove all power from the rig drive system before performing preventive maintenance.

A rig drive system preventive maintenance program should contain the following checks:

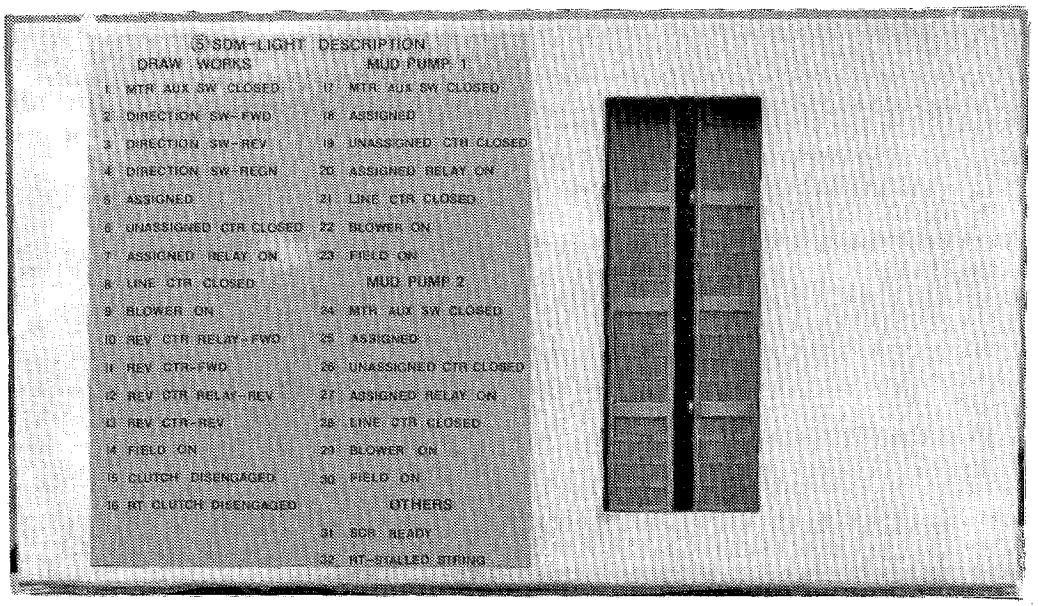
1. Check for loose connections. Vibration may cause terminal board and bus bolt connections to become loose. Loose connections permit moisture and dust accumulations that can result in loss of low level signals at terminal board connections or thermal runaway at bus connections. Problems caused by loose connections can appear and disappear due to vibration.

2. Check for damaged insulation on wires and cables, which can cause intermittent shorting of signal or power cables.

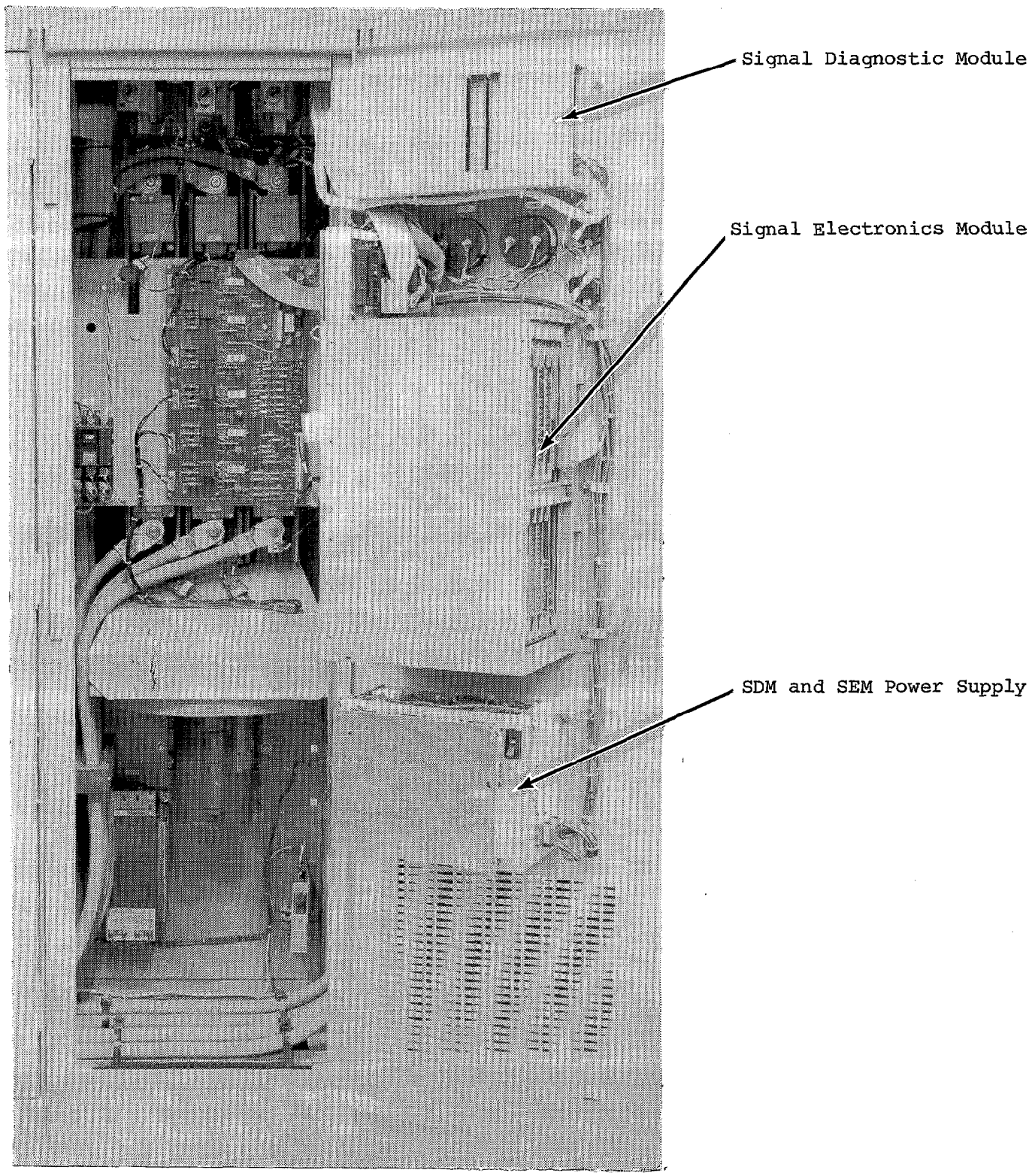
3. Check for dust accumulation on insulation and wire surfaces. Dust accumulating on stand-off insulators can collect enough moisture to produce a current path from bus bars to chassis ground. Dust (especially metallic welding dust) on wire surfaces can cause "tracking" between receptacle pins. Tracking is usually capacitative in nature and involves a build-up of electrical charge along the wire surface, resulting in intermittent problems which are very difficult to locate. Clean bus bars and stand-off insulators with a clean dry cloth (do not use any solvents). Clean wire surfaces between page receptacles by blowing dry air (maximum pressure 80 p.s.i.) across all wire surfaces, directing the air source so that foreign objects are removed rather than relocated.

LOCATING SUBASSEMBLIES

Most subassemblies called out by this troubleshooting guide can be located by referring to the system elementary or u-drill(TM) 1500 Rig Operation Manual (GEK-73911). The SDM or Signal Diagnostic Module (shown below) is referred to repeatedly in this manual and is found inside the Power Conversion Panel (see photo on next page). The SEM or Signal Electronics Module is also found inside the Power Conversion Panel.



Signal Diagnostic Module (SDM)



Inside of Power Conversion Panel

NOTES

SECTION 2
INITIAL PROBLEM DETERMINATION

The first step in troubleshooting the rig drive system is to determine the most applicable problem area (i.e., drilling system problem, motor control center problem, AC motor problem, etc.). This is done by reviewing items 1 through 16 in the table on the next four pages. Once you have determined the most applicable problem area, begin troubleshooting using the listed procedures. The procedures will often refer you to other sections of this manual. Call your General Electric Company representative for assistance if your specific problem area is not covered or if you need help in identifying a problem area.

INITIAL PROBLEM DETERMINATION

Item	Apparent Problem	Action to Take
1	Diesel engine problem	a) Repair any known mechanical problems first. b) Refer to the Power Generation Troubleshooting section of this manual.
2.	AC generator problem	a) Check for excessive vibration, overheating, and sparking at the generator. First try to eliminate mechanical problems. Perform generator insulation checks if possible. b) Refer to the Power Generation Troubleshooting section of this manual.
3.	Generator Synchronizing Problem	NOTE: Manually charge the breaker before trying to synchronize. If the breaker will not charge, refer to the Power Generation Troubleshooting section of this manual. a) Adjust the load bus voltage and frequency for rated values. Remove unnecessary impact load disturbances (such as compressors) while synchronizing. b) Adjust the off line generator voltage to match the load bus voltage (600 volts). c) Adjust the off line generator frequency slightly higher than the load bus frequency. The synchroscope indicator should move slowly in a clockwise direction if the sync switch is in the off line generator position. The sync lights should illuminate each time the synchroscope indicator points at the floor. d) Push the breaker close pushbutton when the sync lights are out and when the synchroscope indicator points vertically at the 12 o'clock position toward the ceiling. e) Refer to equipment operating procedures if adjustments in steps a-d can not be done. f) Replace bad sync light bulbs. Have a qualified electrician replace frequency or synchroscope meters if they fail to respond properly when they are selected. g) Determination of a breaker fault can be made by checking closing coil voltage when the breaker close pushbutton is activated. h) Go to action item 5. j) Go to action item 16.

INITIAL PROBLEM DETERMINATION (CONTINUED)

Item	Apparent Problem	Action to Take
4.	Power limiting problem	a) Check the generator panel meters to see if power limiting occurs at rated engine or generator conditions. It is normal for limiting to occur at rated system limits. b) If limiting occurs before rated limits then: 1) Check the engine air and fuel filters. Check fuel injectors. Check for engine turbocharger overheating. 2) Check ground detection equipment for indication of a system ground fault. 3) Have a qualified person adjust the SCR KVA/KW limits (interface module drawing 173C3316, Sheet 1D). The KW limit serves to prevent overloading the engine. The KVA limits prevents overloading the generator. 4) Check the motor drive for mechanical problems. Mud pumps can limit power if a valve sticks or is broken. 5) Go to action item 12. c) Go to action item 16.
5.	AC generator breaker problem.	a) Refer to the operations manual for instructions. b) Check the power plant shutdown button at the driller's console. It should not be activated. c) Refer to the Power Generation Troubleshooting section of this manual. d) Stop the respective engine, remove all breaker power sources and replace the breaker. e) Go to action item 16.
6.	AC panel problem	a) Refer to the instruction book for handling and storage requirements. b) Refer to the Power Generation Troubleshooting section of this manual. c) Go to action item 16.

INITIAL PROBLEM DETERMINATION (CONTINUED)

Item	Apparent Problem	Action to Take
7.	Power systems grounding problem.	a) Check the ground detection module for the presence of an AC or DC ground fault. 1) If the phase "A","B" or "C" light is out, check the respective light bulb by pushing the Lamp Test button. If the lamp is good, it will light, indicating a ground fault on that phase. Find the ground by eliminating loads as follows: * Turn off drilling assignments (one at a time) until the ground disappears. * Turn off motor control center starters (one at a time) until the ground disappears. * Turn off lighting panel breakers (one at a time) until the ground disappears. * Go to action item 16. 2) If the "POS GND" or "NEG GND" light illuminates, find the DC ground by eliminating SCR assignments one by one until the ground disappears. 3) Go to action item 16.
8.	Motor control center (MCC) problem.	a) If no power is available in an MCC section, check its respective feeder breaker or transformer. b) Check fuses and indicator light bulbs for a failure. c) Check starter armatures for rust and foreign objects. Clean contacts with a crocus cloth and replace any starter contacts that are badly worn. Fuses will blow if the contactor armature is not free to move. d) Refer to starter publications to insure proper rating of breakers and overload heating elements. e) Reset the thermal overload device with the starter reset pushbutton. A 90 to 110% overload device adjustment might be required. This screwdriver adjustment should only be made if the load can withstand up to 110% of the rated overload device current. f) If a starter with automatic starting control does not function when it should, refer to the Drilling Operation Troubleshooting section of this manual. g) Go to action item 16.

INITIAL PROBLEM DETERMINATION (CONTINUED)

Item	Apparent Problem	Action to Take
9.	AC motor problem	a) Repair any known mechanical problems first. b) Check applicable motor control center feeder breakers and starters for proper operation. c) If the motor is started automatically by the SCR control, refer to the Drilling Operation Troubleshooting section of this manual. d) Go to action item 16.
10.	Drilling systems problem.	a) Refer to the Drilling Operation Troubleshooting section of this manual. b) Go to action item 16.
11.	DC motor problem	a) Check for excessive vibration, overheating, and sparking at the motor. Perform field and armature insulation checks if possible. Resolve mechanical problems first. b) Refer to the Drilling Operation Troubleshooting section of this manual.
12.	Motor assignment problem.	a) Try assigning the same motor to another SCR (if possible). Record any variations from the original problem. b) Refer to the Drilling Operation Troubleshooting section of this manual. c) Go to action item 16.
13.	SCR suppression problem.	a) Reset this fault using the NFCB card "UNSUP" switch, located in the Signal Electronics Module (SEM). b) Refer to the Drilling Operation Troubleshooting section of this manual. c) Go to action item 16.
14.	SCR disconnect device problem.	a) Refer to the device publications for operational and maintenance procedures. b) If the breaker can not be charged, closed, or manually tripped, replace it. The 600 volt source must be turned off before performing internal breaker maintenance or before replacing the breaker. c) Go to action item 16.
15.	SCR panel problem.	a) Refer to the instruction book for handling and storage requirements. b) Refer to the Drilling Operation Troubleshooting section of this manual. c) Go to action item 16.

INITIAL PROBLEM DETERMINATION (CONTINUED)

Item	Apparent Problem	Action to Take
16.	Problem has not been covered by items 1 to 15.	a) Complete the following list of questions: 1) What drilling operation was in progress when the problem occurred?_____ 2) Has this problem happened before?_____ If yes: How often?_____ Daytime only?_____ 3) When will the equipment be available for maintenance?_____ 4) Briefly, describe the problem._____ b) Call a representative of General Electric Company for assistance. Give the information in item 16a to the General Electric representative who will service the problem.

SECTION 3
DRILLING OPERATION TROUBLESHOOTING GUIDE

This section contains troubleshooting procedures that relate to drilling operations (i.e., selecting and controlling drilling motors). The troubleshooting procedures are grouped according to the operation being performed and are arranged on separate pages in the following order:

I. DEPENDENT ROTARY OPERATIONS
- 1. Drilling — page 14
- 2. Circulating — page 15
- 3. Make-Up — page 16
- 4. Reaming — page 17
- 5. New Bits — page 18
- 6. Drill Collars — page 19

II. DRAWWORKS OPERATIONS
- 1. Drilling — page 20
- 2. Hoisting — page 21
- 3. Casing — page 22
- 4. Cathead — page 23
- 5. Positioning — page 24
- 6. Mast Raising — page 25
- 7. Spooling — page 26
- 8. Tripping "Clutched" — page 27
- 9. Tripping "Unclutched" — page 28
- 10. Unspooling — page 30
- 11. Slipping Line — page 31

III. MUD PUMP OPERATIONS
- 1. Mud Pump 1 Drilling — page 32
- 2. Mud Pump 2 Drilling — page 33
- 3. Mud Pump 1 Circulating — page 34
- 4. Mud Pump 2 Circulating — page 35
- 5. Mud Pump 1 Cementing — page 36
- 6. Mud Pump 2 Cementing — page 37

Refer to the procedure that best fits the type of operation going on when the problem occurred. You will need to know the SCR motor assignments at the time of the problem and will need to record the status of the SDM lights in the appropriate SCR panel. (NOTE: SDM refers to the Signal Diagnostic Module, located above the assigned SCR Signal Electronics Module (SEM)).

DEPENDENT ROTARY - DRILLING

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR motor blower is on"	Go to next step.	Go to action item 5 (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in forward"	Go to next step.	Go to action item 19 (Drilling Operations)
11	SDM light 3 "Direction switch in reverse"	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD. REGEN"	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to next step.	Go to action item 21 (Drilling Operations)
14	SDM light 11 "Reversing contactor in FWD"	Go to next step.	Go to action item 22 (Drilling Operations)
15	SDM light 12 "Reverse field relay is on"	Go to Action item 23 (Drilling Operations)	Go to next step.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to action item 24 (Drilling Operations)	Go to next step.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling operations)	Go to action item 28 (Drilling Operations)

DEPENDENT ROTARY - CIRCULATING

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5 (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6 (Drilling Operations)
7	SDM light 8 "Armature conatacor closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(If light initially on) Go to next step.	(If light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 Field On"	Go to next step.	Go to action item 9 (Drilling operations)
10	SDM light 2 "Direction switch in forward"	Go to next step.	Go to action item 19 (Drilling operations)
11	SDM light 3 "Direction switch in reverse"	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD. REGEN."	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to next step.	Go to action item 21 (Drilling Operations)
14	SDM light 11 "Reversing contactor in forward"	Go to next step.	Go to action item 22 (Drilling Operations)
15	SDM light 12 "Reverse field relay is on"	Go to action item 23 (Drilling Operations)	Go to next step.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to action item 24 (Drilling Operations)	Go to next step.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 28 (Drilling Operations)	Go to action item 28 (Drilling Operations)

DEPENDENT ROTARY - MAKEUP

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5 (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6 (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	Go to action item 8 (Drilling Operations) (Light always off)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in forward"	Go to next step.	Go to action item 19 (Drilling Operations)
11	SDM light 3 "Direction switch in reverse"	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD. REGEN"	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to next step.	Go to action item 21 (Drilling Operations)
14	SDM light 11 "Reversing contactor in FWD"	Go to next step.	Go to action item 22 (Drilling Operations)
15	SDM light 12 "Reverse field relay is on"	Go to action item 23 (Drilling Operations)	Go to next step.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to action item 24 (Drilling Operations)	Go to next step.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling operations)	Go to action item 28 (Drilling Operations)

DEPENDENT ROTARY - REAMING

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5 (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6 (Drilling Operations)
7	SDM light 8 "Armature contactor closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(If light initially on) Go to next step.	(If light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 Field On"	Go to next step.	Go to action item 9 (Drilling operations)
10	SDM light 2 "Direction switch in forward"	Go to next step.	Go to action item 19 (Drilling Operations)
11	SDM light 3 "Direction switch in reverse	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD. REGEN."	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to next step.	Go to action item 21 (Drilling Operations)
14	SDM light 11 "Reversing contactor in forward"	Go to next step.	Go to action item 22 (Drilling Operations)
15	SDM light 12 "Reverse field relay is on"	Go to action item 23 (Drilling Operations)	Go to next step.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to action item 24 (Drilling Operations)	Go to next step.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling Operations)	Go to action item 28 (Drilling Operations)

DEPENDENT ROTARY - NEW BITS

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5, (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in forward"	Go to action item 10 (Drilling Operations)	Go to next step.
11	SDM light 3 "Direction switch in reverse"	Go to next step.	Go to action item 11 (Drilling Operations)
12	SDM light 4 "Direction switch in FWD. REGEN."	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on	Go to action item 13 (Drilling Operations)	Go to next step.
14	SDM light 11 "Reversing contactor in FWD"	Go to action item 14 (Drilling Operations)	Go to next step.
15	SDM light 12 "Reverse field relay is on"	Go to next step.	Go to action item 15 (Drilling Operations)
16	SDM light 13 "Reversing armature contactor in reverse"	Go to next step.	Go to action item 16 (Drilling Operations)
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling operations)	Go to action item 28 (Drilling Operations)

DEPENDENT ROTARY - DRILL COLLARS

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5 (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6 (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(If light initially on) Go to next step.	(If light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 Field On"	Go to next step.	Go to action item 9 (Drilling operations)
10	SDM light 2 "Direction switch in forward"	Go to action item 10 (Drilling Operations)	Go to next step.
11	SDM light 3 "Direction switch in reverse"	Go to next step.	Go to action item 11 (Drilling Operations)
12	SDM light 4 "Direction switch in FWD. REGEN."	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to action item 13 (Drilling Operations)	Go to next step.
14	SDM light 11 "Reversing contactor in forward"	Go to action item 14 (Drilling Operations)	Go to next step.
15	SDM light 12 "Reverse field relay is on"	Go to next step.	Go to action item 15 (Drilling Operations)
16	SDM light 13 "Reversing armature contactor in reverse"	Go to next step.	Go to action item 16 (Drilling Operations)
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling Operations)	Go to action item 28 (Drilling Operations)

DRAWWORKS - DRILLING

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5, (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in forward"	Go to next step.	Go to action item 19 (Drilling Operations)
11	SDM light 3 "Direction switch in reverse"	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD. REGEN."	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to next step.	Go to action item 21 (Drilling Operations)
14	SDM light 11 "Reversing contactor in FWD	Go to next step.	Go to action item 22 (Drilling Operations)
15	SDM light 12 "Reverse field relay is on"	Go to Action item 23 (Drilling Operations)	Go to next step.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to action item 24 (Drilling Operations)	Go to next step.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling operations)	Go to action item 28 (Drilling Operations)

DRAWWORKS - CASING

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5, (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in forward"	Go to next step.	Go to action item 19 (Drilling Operations)
11	SDM light 3 "Direction switch in reverse"	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD. REGEN."	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to next step.	Go to action item 21 (Drilling Operations)
14	SDM light 11 "Reversing contactor in FWD"	Go to next step.	Go to action item 22 (Drilling Operations)
15	SDM light 12 "Reverse field relay is on"	Go to Action item 23 (Drilling Operations)	Go to next step.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to action item 24 (Drilling Operations)	Go to next step.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling operations)	Go to action item 28 (Drilling Operations)

DRAWWORKS - HOISTING

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5, (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in forward"	Go to next step.	Go to action item 19 (Drilling Operations)
11	SDM light 3 "Direction switch in reverse"	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD REGEN"	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to next step.	Go to action item 21 (Drilling Operations)
14	SDM light 11 "Reversing contactor in FWD."	Go to next step.	Go to action item 22 (Drilling Operations)
15	SDM light 12 "Reverse field relay is on"	Go to action item 23 (Drilling Operations)	Go to next step.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to action item 24 (Drilling Operations)	Go to next step.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling Operations)	Go to action item 28

DRAWWORKS - CATHEAD

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5, (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in forward"	Go to next step.	Go to action item 19 (Drilling Operations)
11	SDM light 3 "Direction switch in reverse"	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD. REGEN."	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to next step.	Go to action item 21 (Drilling Operations)
14	SDM light 11 "Reversing contactor in FWD"	Go to next step.	Go to action item 22 (Drilling Operations)
15	SDM light 12 "Reverse field relay is on"	Go to Action item 23 (Drilling Operations)	Go to next step.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to action item 24 (Drilling Operations)	Go to next step.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling operations)	Go to action item 28 (Drilling Operations)

DRAWWORKS - POSITIONING

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5, (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in forward"	Go to next step.	Go to action item 19 (Drilling Operations)
11	SDM light 3 "Direction switch in reverse"	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD. REGEN"	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to next step.	Go to action item 21 (Drilling Operations)
14	SDM light 11 "Reversing contactor in FWD"	Go to next step.	Go to action item 22 (Drilling Operations)
15	SDM light 12 "Reverse field relay is on"	Go to Action item 23 (Drilling Operations)	Go to next step.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to action item 24 (Drilling Operations)	Go to next step.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling operations)	Go to action item 28 (Drilling Operations)

DRAWWORKS - MAST RAISING

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5, (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in FWD."	Go to next step.	Go to action item 19 (Drilling Operations)
11	SDM light 3 "Direction switch in REV."	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD. REGEN."	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to next step.	Go to action item 21 (Drilling Operations)
14	SDM light 11 "Reversing contactor in FWD"	Go to next step.	Go to action item 22 (Drilling Operations)
15	SDM light 12 "Reverse field relay is on"	Go to Action item 23 (Drilling Operations)	Go to next step.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to action item 24 (Drilling Operations)	Go to next step.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling operations)	Go to action item 28 (Drilling Operations)

DRAWWORKS - SPOOLING

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5, (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in FWD"	Go to next step.	Go to action item 19 (Drilling Operations)
11	SDM light 3 "Direction switch in REV"	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD REGEN"	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to next step.	Go to action item 21 (Drilling Operations)
14	SDM light 11 "Reversing contactor in forward"	Go to next step.	Go to action item 22 (Drilling Operations)
15	SDM light 12 "Reverse field relay is on"	Go to Action item 23 (Drilling Operations)	Go to next step.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to action item 24 (Drilling Operations)	Go to next step.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling operations)	Go to action item 28 (Drilling Operations)

DRAWWORKS - TRIPPING "CLUTCHED"

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5, (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in forward"	Go to next step.	Go to action item 19 (Drilling Operations)
11	SDM light 3 "Direction switch in reverse"	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD. REGEN."	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to next step.	Go to action item 21 (Drilling Operations)
14	SDM light 11 "Reversing contactor in FWD"	Go to next step.	Go to action item 22 (Drilling Operations)
15	SDM light 12 "Reverse field relay is on"	Go to Action item 23 (Drilling Operations)	Go to next step.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to action item 24 (Drilling Operations)	Go to next step.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling operations)	Go to action item 28 (Drilling Operations)

DRAWWORKS - TRIPPING "UNCLUTCHED"

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Draw-works is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5, (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations) This light should be on when SDM lights 10 or 12 are on. It will not be on during regeneration cycle field reversals.
10	SDM light 2 "Direction switch in forward"	Go to action item 10 (Drilling Operations)	Go to next step.
11	SDM light 3 "Direction switch in reverse"	Go to action item 20 (Drilling Operations)	Go to next step.
12	SDM light 4 "Direction switch in FWD. REGEN"	Go to next step.	Go to action item 25 (Drilling Operations)

DRAWWORKS - TRIPPING "UNCLUTCHED" (CONTINUED)

Step	Observe	If Light is On	If Light is Off
13	SDM light 10 "Forward field relay is on"	(Light on only after regeneration) Go to next step.	Go to action item 21 (Drilling Operations) This light should be off during regeneration and then return to on.
14	SDM light 11 "Reversing contactor in FWD"	(Light on only after regeneration) Go to next step.	Go to action item 22 (Drilling Operations) This light should be off during regeneration and then return to on.
15	SDM light 12 "Reverse field relay is on"	(Light on during regeneration) Go to next step.	Go to Action item 26 (Drilling Operations) This light should be on during regeneration and then return to off.
16	SDM light 13 "Reversing armature contactor in reverse"	Go to next step. (Light on during regeneration)	Go to action item 27 (Drilling Operations) This light should be on during regeneration and then return to off.
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 28 (Drilling Operations)	Go to action item 18 (Drilling operations) This light should come on to initialize regeneration. Do not engage the clutch again until regeneration is complete.

DRAWWORKS - UNSPOOLING

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Draw-works is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassign-ed contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5, (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in forward"	Go to action item 10 (Drilling Operations)	Go to next step.
11	SDM light 3 "Direction switch in reverse"	Go to next step.	Go to action item 11 (Drilling Operations)
12	SDM light 4 "Direction switch in FWD. REGEN"	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to action item 13 (Drilling Operations)	Go to next step.
14	SDM light 11 "Reversing contactor in FWD"	Go to action item 14 (Drilling Operations)	Go to next step.
15	SDM light 12 "Reverse field relay is on"	Go to next step.	Go to Action item 15 (Drilling Operations)
16	SDM light 13 "Reversing armature contactor in reverse"	Go to next step.	Go to action item 16 (Drilling Operations) SDM lights 15 and 16
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling operations)	Go to action item 28 (Drilling Operations)

DRAWWORKS - SLIPPING LINE

Step	Observe	If Light is On	If Light is Off
1	SDM light 1 "Motor Auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 5 "Drawworks is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 6 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 7 Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR ready"	Go to next step.	Go to action item 5, (Drilling Operations)
6	SDM light 9 "Motor blower is on"	Go to next step.	Go to action item 6, (Drilling Operations)
7	SDM light 8 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 14 "Field on"	Go to next step.	Go to action item 9 (Drilling Operations)
10	SDM light 2 "Direction switch in forward"	Go to action item 10 (Drilling Operations)	Go to next step.
11	SDM light 3 "Direction switch in reverse"	Go to next step.	Go to action item 11 (Drilling Operations)
12	SDM light 4 "Direction switch in FWD. REGEN"	Go to action item 12 (Drilling Operations)	Go to next step.
13	SDM light 10 "Forward field relay is on"	Go to action item 13 (Drilling Operations)	Go to next step.
14	SDM light 11 "Reversing contactor in FWD"	Go to action item 14 (Drilling Operations)	Go to next step.
15	SDM light 12 "Reverse field relay is on"	Go to next step.	Go to Action item 15 (Drilling Operations)
16	SDM light 13 "Reversing armature contactor in reverse"	Go to next step.	Go to action item 16 (Drilling Operations)
17	SDM lights 15 and 16 "Drawworks and rotary clutches disengaged"	Go to action item 17 (Drilling operations)	Go to action item 28 (Drilling Operations)

MUD PUMP 1 DRILLING

Step	Observe	If Light is On	If Light is Off
1	SDM light 17 "Motor auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 18 "Mud pump 1 is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 19 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 20 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR Ready"	Go to next step.	Go to action item 5 (Drilling Operations)
6	SDM light 22 "Motor blower is on"	Go to next step.	Go to action item 6 (Drilling Operations)
7	SDM light 21 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	Go to next step.	Go to action item 8 (Drilling Operations) (Light always off)
9	SDM light 23 "Field On"	Go to action item 28. (Drilling Operations)	Go to action item 9 (Drilling Operations)

MUD PUMP 2 DRILLING

Step	Observe	If Light is On	If Light is Off
1	SDM light 24 "Motor auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 25 "Mud pump 2 is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 26 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 27 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR Ready"	Go to next step.	Go to action item 5 (Drilling Operations)
6	SDM light 29 "Motor blower is on"	Go to next step.	Go to action item 6 (Drilling Operations)
7	SDM light 28 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 30 "Field On"	Go to action item 28 (Drilling Operations)	Go to action item 9 (Drilling Operations)

MUD PUMP 1 CIRCULATING

Step	Observe	If Light is On	If Light is Off
1	SDM light 17 "Motor auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 18 "Mud pump 1 is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 19 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 20 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR Ready"	Go to next step.	Go to action item 5 (Drilling Operations)
6	SDM light 22 "Motor blower is on"	Go to next step.	Go to action item 6 (Drilling Operations)
7	SDM light 21 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 23 "Field On"	Go to action item 28 (Drilling Operations)	Go to action item 9 (Drilling Operations)

MUD PUMP 2 CIRCULATING

Step	Observe	If Light is On	If Light is Off
1	SDM light 24 "Motor auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 25 "Mud pump 2 is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 26 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 27 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR Ready"	Go to next step.	Go to action item 5 (Drilling Operations)
6	SDM light 29 "Motor blower is on"	Go to next step.	Go to action item 6 (Drilling Operations)
7	SDM light 28 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 30 "Field On"	Go to action item 28 (Drilling Operations)	Go to action item 9 (Drilling Operations)

MUD PUMP 1 CEMENTING

Step	Observe	If Light is On	If Light is Off
1	SDM light 17 "Motor auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 18 "Mud pump 1 is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 19 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 20 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR Ready"	Go to next step.	Go to action item 5 (Drilling Operations)
6	SDM light 22 "Motor blower is on"	Go to next step.	Go to action item 6 (Drilling Operations)
7	SDM light 21 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 23 "Field On"	Go to action item 28 (Drilling Operations)	Go to action item 9 (Drilling Operations)

MUD PUMP 2 CEMENTING

Step	Observe	If Light is On	If Light is Off
1	SDM light 24 "Motor auxiliary switch is closed"	Go to next step.	Go to action item 1 (Drilling Operations)
2	SDM light 25 "Mud pump 2 is assigned"	Go to next step.	Go to action item 2 (Drilling Operations)
3	SDM light 26 "An unassigned contactor is closed"	Go to action item 3 (Drilling Operations)	Go to next step.
4	SDM light 27 "Assignment relay is on"	Go to next step.	Go to action item 4 (Drilling Operations)
5	SDM light 31 "SCR Ready"	Go to next step.	Go to action item 5 (Drilling Operations)
6	SDM light 29 "Motor blower is on"	Go to next step.	Go to action item 6 (Drilling Operations)
7	SDM light 28 "Armature contactor is closed"	Go to next step.	Go to action item 7 (Drilling Operations)
8	SDM light 32 "Throttle off"	(Light initially on) Go to next step.	(Light always off) Go to action item 8 (Drilling Operations)
9	SDM light 30 "Field On"	Go to action item 28 (Drilling Operations)	Go to action item 9 (Drilling Operations)

ACTION ITEMS - DRILLING OPERATIONS

Item	Apparent Problem	Action to Take
1.	Motor auxiliary switch light will not come on. (Light bulb is OK).	a) Pull the auxiliary switch sliding retainer bar to release the spring loaded pushbutton on the side of the motor. b) Check the switch for mechanical binding. c) Check motor control cable interconnections - motor terminals "A" to "C" (sheets 0D - 0G of the systems elementary).
2.	Console motor assignment will not activate. (Light bulb is OK).	a) Make a physical inspection of the assignment switch. Check for binding of contacts and loose connections. b) Check driller's console DC1 cable interconnections (sheet 1A of the systems elementary).
3.	Unassigned contactor is closed. (Light bulb is OK)	a) Remove contactor power by opening the applicable SCR disconnect switch. If the contactor has welded contacts, replace all damaged contactor parts. b) Repeat step 2 action item if the motor remains assigned with all assignment switches off. c) Check contactor auxiliary interlocks for mechanical binding and electrical shorts (see sheet 1 of contactor module drawing).
4.	Assignment relay is not energized. (Light bulb is OK)	a) Check motor blower and armature contactor circuits which are energized by this relay. Check circuit power sources for an output. b) Check relay receptacle seating and relay coil voltage (sheet 1B of the interface module drawing).
5.	SCR READY light does not come on. (Light bulb is OK)	a) Check PS1B power supply output levels at NFCB card test ring points. Yellow "IMOK" lights should be illuminated on SEM cards (NOWA and NOWB, if used). b) Replace the power supply if no "IMOK" lights illuminate when 120 volts AC is applied to the PS1B (two red twisted pair wires on the PS1B terminal board). c) Replace any card which will not reset to illuminate it's own "IMOK" light when voltage at NFCB test points are within the following tolerances (measured to ACOM): ACOM ANALOG COMMON TP12 P5 (+)5 ±.05 VDC TP10 P15 (+)15 ±.05 VDC TP5 N15 (-)15 ±.05 VDC TP4 P28 (+)28 ±0.7 VDC TP7 NOTE: Refer to PS1B module documentation for adjustment data.

ACTION ITEMS - DRILLING OPERATIONS (CONTINUED)

Item	Apparent Problem	Action to Take
		d) Check power converter blower starter auxiliary contact. Start the blower or reset the starter overload if the blower will not start (See sheet 4A of the power converter panel drawing). e) Check the "HEAT SINK OVERTEMP" device by checking circuit continuity on the device side of plug P2 (sheet 4A of the power converter panel drawing). f) Reset "BLOWER LOSS" or "SUPPRESSION" lights using the NFCB SRST and TUNSU switches. This action should also cause the PLL light on the NFCB card to illuminate. Swap NFCB cards (keep all daughter adjustment boards with the original unit) if faults do not reset.
6.	MOTOR BLOWER light does not illuminate at the SDM (Light bulb is OK)	a) If the motor blower light is illuminated at the driller's console, check the 51 pin ribbon wiring (receptacle seating) between the interface module and the signal diagnostic modules. b) Check motor starter for "automatic" operation mode on starter selector switch. c) Check motor starter for a thermal overload trip. Reset as required. d) Check motor starter 120 volt control power and fuses. e) Check wiring between the starter and the interface module if light 7 is on at the signal diagnostic module.
7.	Armature contactor does not close. (Light bulb is OK)	a) When no assignments are made, verify that no contactors are picked up. Free any contacts which may be welded (a welded contact should be investigated by a representative of General Electric). Investigate interface module wiring if a source is energizing an unassigned contactor. b) Remove contactor power and check for mechanical binding. c) Check wiring between the contactor and the interface module if light 7 is on at the signal diagnostic module (SDM).
8.	Throttle off light does not come on when the throttle is off and the motor is assigned. (Light bulb is OK)	a) Check throttle handle screws to insure that no slip occurs. b) Check throttle miniature switch and activating throttle cam for proper alignment. c) Check interconnections between the console throttle and the interface module. d) Check interconnections between the interface module and the signal diagnostic module.

ACTION ITEMS - DRILLING OPERATIONS (CONTINUED)

Item	Apparent Problem	Action to Take
9.	"FIELD ON" light does not illuminate. (Lamp is OK)	a) If the motor has a series field, this light should be on if 28 volts DC is available. b) If the motor has a shunt field: 1) Check the field exciter for a power output. Check for a short in the field circuit before resetting the field supply circuit breaker if it has tripped. 2) Check for 150 volts (RMS) between each phase and transformer center-tap input to the field exciter. 3) If the exciter output is greater than 50 volts DC check for an open in the field circuit. Check for grounds (disconnect the exciter before performing meggar checks). 4) If field current is greater than 8 amps, replace the exciter. 5) Check interconnections between the field exciter and the interface module. 6) Check interconnections between the interface module and the signal diagnostic module.
10.	"DIRECTION SWITCH IN FORWARD" light is on when direction switch is in another position.	a) Check interconnections between the driller's console direction switch and the interface module. b) Check interconnections between the interface module and the signal diagnostic module. c) Check the direction switch for a foreign object and a short circuit
11.	"DIRECTION SWITCH IN REVERSE" light is off when DIRECTION switch is in reverse. (Lamp is OK)	a) Perform action items 10a and 10b. b) Check the direction switch for a foreign object, loose connections, or an open circuit.
12.	"DIRECTION SWITCH IN FWD. REGEN." light is on when DIRECTION switch is in another position.	a) Perform action items 10a - 10c.
13.	"FORWARD FIELD RELAY IS ON" light is on when the DIRECTION switch is in another position.	a) Perform action items 10C and 15. b) Replace relay FDDX in the interface module. c) Check for foreign objects and shorted receptacle pins on the interface module.
14.	"REVERSING CONTACTOR IN FORWARD" light is on when the DIRECTION switch is in reverse.	a) If the motor has a shunt field, this light has no significant purpose. Perform action items 13a - 13c. b) If the motor has a series field: 1) Perform action items 13a - 13c. 2) Perform action items 3a and 3c for the reversing contactor.

ACTION ITEMS - DRILLING OPERATIONS (CONTINUED)

Item	Apparent Problem	Action to Take
15.	"REVERSE FIELD RELAY IS ON" light is off when the DIRECTION switch is in reverse. (Light bulb is OK)	a) Perform action items 11a - 11b. b) Replace relay RDDX if the relay was properly seated. c) Check SEM jack JG interconnections to the signal diagnostic module. d) Perform action item 28.
16.	"REVERSING ARMATURE CONTACTOR IN REVERSE" light is off when the DIRECTION switch is in reverse.	a) Perform steps 10a, 10b, 11b, 13c and 14b2 if the motor has a series field. Motors with shunt field controls do not use this light. b) Check the contactor for mechanical binding.
17.	"DRAWWORKS AND ROTARY CLUTCH DISENGAGED" lights are on when the clutch is engaged.	a) Check the air pressure at the switch for at least 90 PSI when the clutch is engaged. b) Check for foreign objects in the switch air line. c) Check for 28 volts DC between driller's console wires: 1) 1DC1 ands 1DC11 when the drawworks clutches (high and low drums) are disengaged. 2) 1D1 and 1D12 when the rotary clutch is disengaged. d) The voltages in step C should go from 28 volts to 0 volts when respective clutches are engaged. e) Check interconnections between the pressure switches and the interface module. f) Check interconnections between the interface module and the signal diagnostic module.
18.	"DRAWWORKS AND ROTARY CLUTCH DISENGAGED" lights are off when the clutch is disengaged. (Light bulb is OK)	a) Perform steps 17c, 17d, 17e and 17f. b) Verify loss of pressure switch air pressure when the clutch is disengaged.
19.	"DIRECTION SWITCH IN forward" light is on when direction switch is in another position.	a) Perform action items 11a and 11b.
20.	"DIRECTION SWITCH IN REVERSE" light is on when direction switch is in another position.	a) Perform action items 10a, 10b ands 10c.
21.	"FORWARD FIELD RELAY IS ON" light is off when the DIRECTION switch is in FORWARD or forward regeneration and an assignment has been made. (Light bulb is OK)	a) Perform action item 19a. b) Replace relay FDDX if the relay was properly seated. c) Check SEM jack JG interconnections to the signal diagnostic module. d) Perform action item 28.

ACTION ITEMS - DRILLING OPERATIONS (CONTINUED)

Item	Apparent Problem	Action to Take
22.	"REVERSING CONTACTOR IN FORWARD" light is off when the DIRECTION switch is in FORWARD/FWD. REGEN. and an assignment has been made. (Light bulb is OK).	a) Perform action items 16a and 16b.
23.	"REVERSE FIELD RELAY IS ON" light is on when the DIRECTION switch is in another position.	a) Perform action items 10c and 21.
24.	"REVERSING ARMATURE CONTACTOR IN REVERSE" light is on when the DIRECTION switch is in another position.	a) If the motor has a shunt field, this light has no significant purpose. Perform action item 23a.
25.	"DIRECTION SWITH IN FWD. REGEN." is off when the DIRECTION switch is in FORWARD REGENERATION. (Light bulb is OK).	a) Perform action items 11a and 11b.
26.	"REVERSE FIELD RELAY IS ON" light does not come on while truing to regenerate. (Light bulb is OK)	a) If the motor speed was not at least 12% higher than the cathead speed, regeneration is not required and the control will not automatically initiate regeneration. b) Check drawworks reversing by placing the DIRECTION switch in reverse. If SDM light 12 does not illuminate, perform action item 15. c) Replace the NOWA card in SEM slot B. Do not exchange the "daughter board" (DS3800DOWA). d) Replace the HLEA card in SEM slot 2C. e) Replace the NGPA card in SEM slot 1E.
27.	"REVERSING ARMATURE CONTACTOR IN REVERSE" light does not come on when trying to regenerate. (Light bulb is OK).	a) Perform action items 16a, 16b and 26.

ACTION ITEMS - DRILLING OPERATIONS (CONTINUED)

Item	Apparent Problem	Action to Take
28.	Signal diagnostic module lights are OK, but the drive does not work properly.	a) Check DS3800NHVB lights with SCR bridge power on: 1) If any of the 6 yellow lights are out: -Check SCR fuse indicators for a bad fuse. -Check for a shorted SCR. 2) If any of the 6 red lights are out while operating: -Replace the DS3800NFCB card. -Replace the DS3800NHVB card. b) * Replace or swap the following cards in the order listed: ALL JOBS - 1) DS3800NOWA 2) DS3800NFCB 3) DD3800HLEA (Row 2 - slot E) 4) DS3800HLEA (Row 2 - slot F) 5) DS3800HLEA (Row 2 - Slot D) 6) DS3800HLEA (Row 2 - Slot C) 7) DS3800NGPA (Row 1 - Slot E) SERIES MOTOR JOBS - 8) DS3800NOWB 9) DS3800COWC (This card has adjustments which might require recalibaration if replaced.) ALL JOBS - INTERFACE MODULE - 10) IC3600AOAL1 (Row 1 - Slot E) (This card has adjustments which might require recalibration if replaced.) c) Call a representative of General Electric Company. Have the following information available in order to expedite repairs: 1) Type of operational problem. 2) Frequency of problem. 3) Status of SDM lights 4) Status of red light on NOWA cards when problem occurred. 5) Status of red and yellow lights on the NHVB card of the problem SCR drive when the problem occurred.

* NOTE: When swapping or replacing cards --

1. Do not remove or insert cards with power on.
2. Do not swap or replace daughter boards (plug-in adjustment cards).
3. Do not replace a card with one which has a lower revision part number.

NOTES

SECTION 4
POWER GENERATION TROUBLESHOOTING GUIDE

This section contains troubleshooting procedures that relate to power generation problems. The troubleshooting procedures are grouped in five tables according to the position of the engine control switch when the problem occurred:

1. Engine control switch in OFF position
2. Engine control switch in IDLE position
3. Engine control switch in RUN position
4. Engine control switch in LOAD position (one generator on line)
5. Engine control switch in LOAD position (multiple generators on line)

By using the applicable table on the next two pages, you will be guided to an action item located later in the manual which will tell you what to do to solve the problem.

All lights referred to in this section of the manual are located on the AC Engine/Generator Control Panel, also known as the Generator Electronics Module or "GEM". When reading the lights, a flashing light should be interpreted as off. Always refer to the action item related to the flashing light before referring to action items for lights that are simply off.

1 - ENGINE CONTROL SWITCH IS IN "OFF"

Step	Question	Yes	No
1	Is this an engine starting problem?	Go to action item 1.	Go to next step.
2	Is the PWR SUPPLY OK light on?	Go to action item 9.	Go to action item 2.

2 - ENGINE CONTROL SWITCH IS IN "IDLE"

Step	Question	Yes	No
1	Is this an engine starting problem?	Go to action item 1.	Go to next step.
2	Is the PWR SUPPLY OK light on?	Go to next step.	Go to action item 2.
3	Is the UNDERVOLT light on?	Go to next step.	Go to action item 3.
4	Is the NO TACH LOSS light on?	Go to next step.	Go to action item 4.
5	Is the NO OVRSPD light on?	Go to next step.	Go to action item 5.
6	Is the NO OV/UF light on?	Go to next step.	Go to action item 6.
7	Is the NO REV PWR light on?	Go to action item 9.	Go to action item 7.

3 - ENGINE CONTROL SWITCH IS IN "RUN"

Step	Question	Yes	No
1	Is this an engine starting problem?	Go to action item 1.	Go to next step.
2	Is the PWR SUPPLY OK light on?	Go to next step.	Go to action item 2.
3	Is the UNDERVOLT light on?	Go to next step.	Go to action item 3.
4	Is the NO TACH LOSS light on?	Go to next step.	Go to action item 4.
5	Is the NO OVRSPD light on?	Go to next step.	Go to action item 5.
6	Is the NO OV/UF light on?	Go to next step.	Go to action item 6.
7	Is the NO REV PWR light on?	Go to action item 9.	Go to action item 7.
8	Is the GEN ON light on when the breaker is closed?	Go to action item 9.	Go to action item 8.

4 - ENGINE CONTROL SWITCH IS IN "LOAD" (ONE GENERATOR ON LINE)

Step	Question	Yes	No
1	Is this an engine starting problem?	Go to action item 1.	Go to next step.
2	Is the PWR SUPPLY OK light on?	Go to next step.	Go to action item 2.
3	Is the UNDERVOLT light on?	Go to next step.	Go to action item 3.
4	Is the NO TACH LOSS light on?	Go to next step.	Go to action item 4.
5	Is the NO OVRSPD light on?	Go to next step.	Go to action item 5.
6	Is the NO OV/UF light on?	Go to next step.	Go to action item 6.
7	Is the GEN ON light on when the breaker is closed?	Go to action item 9.	Go to action item 8.

5 - ENGINE CONTROL SWITCH IS IN "LOAD" (MULTIPLE GENERATORS ON LINE)

Step	Question	Yes	No
1	Is this an engine starting problem?	Go to action item 1.	Go to next step.
2	Is the PWR SUPPLY OK light on?	Go to next step.	Go to action item 2.
3	Is the UNDERVOLT light on?	Go to next step.	Go to action item 3.
4	Is the NO TACH LOSS light on?	Go to next step.	Go to action item 4.
5	Is the NO OVRSPD light on?	Go to next step.	Go to action item 5.
6	Is the NO OV/UF light on?	Go to next step.	Go to action item 6.
7	Is the NO REV PWR light on?	Go to next step.	Go to action item 7.
8	Is the GEN ON light on when the breaker is closed?	Go to action item 9.	Go to action item 8.

ACTION ITEMS - POWER GENERATION

Item	Apparent Problem	Action to Take
1	Engine starting problem.	a) Check the engine control switch. It must be in the "IDLE" position for engine starting. b) Check air starting pressure for adequate starting power. c) Check for air intake and possible fuel line restrictions. Refer to the prime mover maintenance manual for other possible mechanical problems. d) If the generator - engine module (1GEM) "PWR. SUPPLY OK" light IS ON AND IF THE CONTROL SWITCH IS IN "IDLE", check the following: 1) Disconnect the actuator power plug and verify a power input from the GEM while cranking the engine. Typical voltage while cranking is above 5 volts DC. If voltage is present with the proper polarity, an actuator or engine problem exists. 2) If step 1 voltage is not present, check for actuator power at the GEM terminal board points 15 (+) to 16 (-) [count left to right] for 5 or more volts while cranking the engine. If voltage is present, an interconnection problem exists between the GEM and the actuator. If voltage is not present, replace the GEM.
2	"PWR SUPPLY OK" light is on. NOTE: All GEM lights are normally extinguished if the engine control switch is in the "OFF" position.	a) Check for a 12 volt "cold start battery" output at the cold start battery module. The module light should illuminate if power is available. A minimum of 10 volts is required for proper GEM operation. If the battery will not charge, replace the cold start battery module. b) Replace the generator-engine module if 12 volt power is available and the engine control switch is not off. c) Connect a 12 to 60 volt DC battery (8 amp-hour minimum) to GEM terminals 1 (+) and 2 (-) on terminal board CTB. Facing the mounted GEM backplane, terminal 1 is in the first left screw slot position; terminal 2 is just to the right of terminal 1.

ACTION ITEMS - POWER GENERATION (CONTINUED)

Item	Apparent Problem	Action to Take
3	"UNDERVOLT" light is not on.	a) This light should not illuminate until the generator line to line voltage exceeds 450 volts. b) If the panel volttmeter indicates a voltage above 500 volts and the "UNDERVOLT" light is still out, replace the GEM.
4	"NO TACH LOSS" light is not on. NOTE: Proceed with "action" to be taken" steps if the problem occurs again after pushing the GEM reset pushbutton.	a) Check the engine magnetic pickup for an input to the GEM. Typical input resistance is between 100 and 300 ohms. Typical inpout voltage is between 5 and 50 volts at rated speed. Adjust, clean, or replace the pickup if interconnections check good. b) Configure for tachless operation by moving the blue "berg jumper" (J1) on the GEM backplane to its alternate position (there are 3 pins labeled J1. The unused to center pin is the alternate position). Tachless operation is to be used only if a spare module is not available for replacement. Replace the GEM.
5	"No OVRSPD" light is not on.	a) This is a status light which indicates an engine overspeed condition (1.08 times rated speed). This condition probably occurred due to a loss of speed feedback or a change in loading condiditions. 1) If the "NO TACH LOSS" light is also out, repeat action item 4. 2) Push the GEM reset pushbutton. Put more engines on line if load changes are causing the overspeed trip. b) Replace the GEM. Repeat step 4B if no GEM replacement is available.
6	"NO OV/UF" light is not on.	a) Push the GEM reset pushbutton and try again. This condition may occur if the tach thumbwheel switch is in a marginal position. Only qualified personnel should attempt module adjustments. Call a General Electric representative if these trips persist. Do not attempt to reset if an overvoltage condition does exist since this could damage the generator. The voltage at rated speed should be regulated to not exceed 600 volts ±10%. b) Replace the GEM. Repeat step 4B if no GEM replacement is available.

ACTION ITEMS - POWER GENERATION (CONTINUED)

Item	Apparent Problem	Action to Take
7	"NO REV PWR" light is not on.	a) This is a status light which indicates a load imbalance between generators. An imbalance is more significant if two units are on line than when more units are on-line and a sudden load change is encountered. 1) Increase the number of on-line generators to better accommodate peak loading conditions. 2) Use the speed trim adjustment to balance the KW load and to maintain rated system frequency. 3) Have systems power load sharing adjusted by a qualified representative of General Electric. Return all adjustments to their original startup position. 4) If the light will not reset, replace the GEM.
8	"GEN ON" light is not on when the generator breaker is closed.	a) Replace the GEM. b) Check the generator breaker auxiliary interlock which is wired to GEM jack JSYSA pins 4 to 5. There should be a continuity between pins 4 and 5 when the breaker is open. There should not be a continuity when the breaker is closed. NOTE: An open circuit would cause the light to be on at all times. 1) Check interconnections from JSYSA and the breaker. 2) Replace the breaker if interconnections are good but no continuity exists between JSYSA (4) and JSYSA (5) when the breaker is open.
9	Power generation systems "don't work".	NOTE: There are many electrical and mechanical problems which are unique to our customer's systems. At this point it is wise to call a representative of General Electric who can analyze your specific systems.

CUSTOMER CHECKLIST FOR UNRESOLVED PROBLEMS

If a problem cannot be solved and you must call General Electric for service, be sure to fill out this checklist and have it available for the service engineer. The information on this checklist will help to expedite repairs.

1) Description of problem: _____

_____

_____

_____

_____

_____

_____

2) What drilling operation was underway at the time of failure? _____

_____

_____

3) Frequency of problem: ___ Steady ___ Intermittent (____ times per day)

4) SDM lights: (cross out lights which are ON when the problem occurs):

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32

5) NOWA red light ___ ON ___ OFF

6) NHVB lights: Yellow ___ ON ___ OFF

Red ___ ON ___ OFF

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